Curriculum Vitae

Dr. Rachel Glenn

Medical Physics Fellow
Department of Imaging Physics
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EDUCATION

Res. **Medical Physics Fellow (Resident)**
Department of Imaging Physics, UT MD Anderson Cancer Center / UTHealth Houston
2025–Present (Expected Completion: August 2028)
ABR Eligible, passed ABR 1. TMB License: TMP02001022

M.S. **Medical Physics**
Department of Imaging, UTHealth / MD Anderson, 2024
Thesis: Hybrid Quantum/Classical Image Segmentation Algorithms
Committee: David T. Fuentes (chair), Gary V. Martinez, Jason Stafford, James A. Bankson

Ph.D. **Condensed Matter Physics**
Department of Physics, University of Utah, Salt Lake City, 2012
Thesis: Many-Body and Spin-Orbit Aspects of the Alternating Current Phenomena
Committee: M.E. Raikh (chair), Oleg Starykh, Christoph Boehme, Kyle Dawson, Elena Cherkaev

B.S. **Physics with Honors, Minor in Mathematics**
Weber State University, Ogden, 2008

B.S. **Electronics Engineering Technology**
Weber State University, Ogden, 2008

A.S. **Electronics Technology**
Salt Lake Community College, Salt Lake City, 2002

TEACHING

2026 **Imaging & Innovation Summer Immersion Program**
Assisted in Nuclear Medicine Lab, UTHealth / MD Anderson

2020 **Electronics Laboratory**

Taught Lab course as Adjunct Professor, Department of Electrical and Computer Engineering, Baylor University

2015 **Special Topics in Physical Chemistry (CEM-988)**
Developed entire course, taught students and graded all homework. Postdoctoral Scholar, Department of Chemistry, Michigan State University,

RESEARCH

2025- present **Medical Physics Fellow Resident**
Department of Imaging, UTHealth/MD Anderson

- Wrote code and manuscript, literature review, broadened dataset inclusion and models working with patient images, for translational research in quantitative MRI for precision oncology, focusing on diffusion MRI, multiparametric imaging, and biopsy-correlated radiomic biomarkers for glioma characterization.
- Developed dataset and validated imaging biomarkers (ADC, IVIM, diffusion kurtosis, CTRW) and machine learning models to predict tumor cellularity and support image-guided diagnosis, adaptive radiotherapy, and treatment planning.
- Integrate quantitative imaging, pathology, and computational modeling to improve clinically interpretable biomarkers for MRI-guided cancer care.
- Assisting with modifying diffusion protocols for advancing quantitative MRI, including oscillating gradients. Understanding how to modify current sequences from a laptop connected to a GE scanner.

2024-2025 **Associate Fellow / Senior AI & Machine Learning Engineer**
Lockheed Martin

- Associate Fellow requires nomination by a Senior Fellow, Bryan Bell. It requires demonstration of leadership abilities, communication, deep subject-matter expertise, driving research, mentoring, and publishing.
- Researched, developed, and implemented an advanced computational imaging algorithms for deployable synthetic aperture radar (SAR) systems, including image reconstruction, motion compensation, compressed sensing, and three-dimensional image formation. In addition, located needed dataset for development and analysis of algorithms.
- Designed AI-assisted reconstruction and image enhancement methods to improve image quality, robustness, and operational performance in challenging sensing environments.
- Collaborated with multidisciplinary engineering teams to translate novel computational imaging algorithms into deployable real-time imaging systems and collect data to test image reconstruction algorithm development.
- Authored multiple internal research proposals in quantum computing for autonomous sensing systems, securing funding for next-generation computational imaging research. We were awarded seedling funding.
- Mentored rising technical staff in quantum computing. Developed a self-study

guide for engineers interested in quantum computing and pushed for management to provide training funding in quantum AI for AI engineers.
- Part-time consultant during 2021-2024.

2021-2024 **Research Associate / Graduate Student**
UTHealth / MD Anderson
- Assisted in improving deep learning and image segmentation methods for MRI-based radiation therapy planning and quantitative image analysis.
- Implemented and analyzed classical and quantum-inspired computational approaches for medical image segmentation, auto-contouring, and treatment planning.
- Contributed to the development and validation of MRI segmentation models for liver and abdominal imaging across diverse clinical datasets.

2017-2021 **Artificial Intelligence Architect in Research and Development**
Advanced Development Programs, Lockheed Martin
- Contributed to the technical leadership of multidisciplinary research in artificial intelligence, computational modeling, and advanced sensing technologies.
- Managed international collaborations with universities, industry, and government to develop research programs in quantum technologies and advanced computing. Explained/Taught the latest research, advantages, and limitations to key-stake holders and upper management. Reviewed proposals from international quantum computing and devices for feasibility and investment.
- Developed digital twin simulations and AI-enabled 3D environmental models for autonomous systems.
- Authored successful research proposals that secured funding for next-generation AI acceleration technologies.

2016-2017 **Senior Research Associate: Daniel Crawford**
Department of Chemistry, Virginia Polytechnic Institute and State University
- Collaborated with Quantum Chemistry research group to merge my background research in time-dependent spectroscopy with quantum chemistry to design and implemented time-dependent quantum chemistry methods for modeling molecular response and spectroscopy.
- Designed/Implemented computational algorithms for simulating light–matter interactions and electronic structure.

2014-2016 **Postdoctoral Research Position: Marcos Dantus**
Department of Chemistry, Michigan State University
- Theoretical physicists now employed with collaborator to develop theoretical models of their experimental measurements of molecular dynamics. The modality used was multidimensional optical spectroscopy, both quantum and classical electromagnetic fields.
- Went into the laboratory to assist, understand, and model molecular dynamics measured and observed in optical spectroscopy.

- Required to continuously apply to all funding opportunities to support myself.

2013-2014 **Postdoctoral Research Position: Shaul Mukamel**
Department of Chemistry, University of California, Irvine
- Learned the nonlinear optical spectroscopy formalism, which is based upon Quantum Field Theory. Then developed new theoretical and computational methods and predictions for nonlinear optical spectroscopy. One theoretical prediction was later verified by a Nobile-prize winner with the Frequency Comb Spectroscopy, Theodor W. Hänsch.
- Studied pulse design, timing, and nonlinear signal–matter interactions to understand how excitation parameters shape spectroscopic imaging signals and reveal molecular dynamics—skills transferable to MRI pulse-sequence development.
- Interpreted experimental imaging and spectroscopy from the Dantus research group to develop theoretical models to reproduce the molecular dynamics within complex multidimensional optical measurements.

2012 **Postdoctoral Research Position: M. E. Raikh**
Department of Physics and Astronomy, University of Utah
- Based upon graduate research with modeling spin dynamics, investigated magnetic resonance phenomena of spin dynamics with magnetic field modulations.
- Assisted with and published theoretical models describing magnetic resonance response in condensed matter systems.
- Based upon prior machine-shop experience, designed and built probe & coils to reproduce theoretical predictions and use in magnetic bore. Sought assistance when needed.
- These skills are transferable to developing theory/experiments of spin dynamics for the use of gradients in MRI.

2010-2012 **Graduate Research Assistant: Oleg Starykh**
Department of Physics and Astronomy, University of Utah
- Learned theoretical condensed matter physics from Russian Physicists.
- Assisted with theoretical development of describing experimental results in spin-dependent transport, magnetic resonance, and light–matter interactions in semiconductor materials. Some papers were purely theoretical and unproven experimentally. Developed theoretical models of systems included describing electrically detected magnetic resonance (EDMR), spin-orbit interactions, and quantum transport.

IMAGING & AUTOMATION SYSTEMS EXPERIENCE

2002-2012 **Airways Transportation System Specialist (ATSS), FV-H (2101)**
Interim appointment as Unix Administrator for STARS automation system (FV-I)
Federal Aviation Administration, Salt Lake City, UT
- Earned journeyman certifications (4 years completion time) on the Mode-S, ASR-9, ASDE-X, TDWR, LLWAS, and STARS radar and automation platforms.

- Maintained, calibrated, upgraded, and certified mission-critical radar and automation systems operating in a 24/7 safety-critical environment.
- Performed system commissioning, acceptance testing, preventive maintenance, troubleshooting, and performance verification of complex electronic and imaging-related systems.
- Promoted temporarily as Admin of STARS. This system processes multiple source radar and weather data, mosaics, and displays all of air traffic control user stations, including flight plans, warnings, and user setting, for supporting real-time operational control of the airspace.
- Developed technical expertise in quality assurance, system reliability, fault analysis, documentation, and regulatory compliance.
- Collaborated with engineers, operators, vendors, contractors, and management (both my own and air traffic control) to minimize downtime and maintain continuous system availability.
- Trained and certified technical staff through structured on-the-job training and competency evaluations.
- Majority voted in as Union Representative for several years.

INNOVATION & PROFESSIONAL DEVELOPMENT

- MIT Innovation Initiative Hackathon, MIT, Cambridge, MA (2019)
- LM Bright-Ideas (Sharktank) seedling funding AI deployed to FPGA (2019)
- LM Bright-Ideas (Sharktank) seedling funding atmostpheric simulations (2020)
- MIT Silicon Photonics Workshop, MIT, Cambridge, MA (2020)
- LM Bright-Ideas (Sharktank) seedling funding, Quantum Reinforcement Learning (2025)
- Communication with Complex People, Self-study (2023-2026)
- Theoretical predicted how to control of chemical bonds, though light matter interaction, still unproven experimentally (2022 patient).

PATENTS

2022 Method of Single Pulse Detection of the Real and Imaginary Components of the Nonlinear Susceptibility to Detect Molecular Dynamics via the Geometric Phase Effect, Publication No. 20200003681.

PUBLICATIONS

2026 Rachel Glenn, Tucker Netherton, Adrian Celaya, Mohamed Eltaher, Beatrice Riviere, Eugene J. Koay, David Fuentes. *Evaluation of quantum contouring algorithms for treatment planning on MR abdominal images*. Frontiers in Oncology. 2025;15:1553539. doi:10.3389/fonc.2025.1553539. Included in the 2026 Frontiers Research Topic, Emerging Fast Medical Imaging Techniques in Radiology.

2026 Rachel Glenn, Rajarajeswari Muthusivarajan, David A. Hormuth II, Max Wintermark,

Ken Hwang, Dong-Eog Kim, David Fuentes, Dawid Schellingerhout. *Apparent Diffusion Coefficient (ADC) as a single predictor is insufficient to predict cellular density.* European Journal of Radiology 204 113119

2025 Rachel Glenn, Tucker Netherton, Adrian Celaya, Mohamed Eltaher, Beatrice Riviere, Eugene J. Koay, David Fuentes. *Evaluation of Quantum Contouring Algorithms for Treatment Planning on MR Abdominal Images. Frontiers in Oncology* 15, 1553539.

2024 Awj Twam, Adrian E. Celaya, Megan C. Jacobsen, Rachel Glenn, Peng Wei, Jia Sun, Ann Klopp, Aradhana M. Venkatesan, David Fuentes, *Two-Stage Segmentation of Cervical Tumors Using PocketNet*. arXiv:2409.11456.

2024 Nihil Patel, Adrian Celaya, Mohamed Eltaher, Rachel Glenn, Kari Brewer Savannah, Kristy K. Brock, Jessica I. Sanchez, Tiffany L. Calderone, Darrel Cleere, Ahmed Elsaiey, Matthew Cagley, Nakul Gupta, David Victor, Laura Beretta, Eugene J. Koay, Tucker J. Netherton, and David T. Fuentes, *Training Robust T1-Weighted Magnetic Resonance Imaging Liver Segmentation Models Using Ensembles of Datasets with Different Contrast Protocols and Liver Disease Etiologies*, *Scientific Reports* 14, 20988.

2024 Adrian Celaya, Evan Lim, Rachel Glenn, Brayden Mi, Alex Balsells, Dawid Schellingerhout, Tucker Netherton, Caroline Chung, Beatrice Riviere, David Fuentes, *MIST: A Simple and Scalable End-to-End 3D Medical Imaging Segmentation Framework*, arXiv preprint arXiv:2407.21343.

2016 Ilyas Saytashev, Rachel Glenn, Galina A. Murashova, Saman Osseiran, David Spence, Cynthia L. Evans, and Marcos Dantus, *Multiphoton Excited Hemoglobin Fluorescence and Third Harmonic Generation for Non-Invasive Microscopy of Stored Blood*, *Biomed. Opt. Express* 7, 3449–3460.

2016 Rachel Glenn and Marcos Dantus, *Molecular Level Crossing and the Geometric Phase Effect from the Optical Hanle Perspective*, *Phys. Rev. A* 93, 043402.

2016 Rachel Glenn and Marcos Dantus, *Single Broadband Phase-Shaped Pulse Stimulated Raman Spectroscopy for Standoff Trace Explosive Detection*, *J. Phys. Chem. Lett.* 7, 117–125.

2015 Richa Mittal, Rachel Glenn, Ilyas Saytashev, Vadim V. Lozovoy, and Marcos Dantus, *Femtosecond Nanoplasmonic Dephasing of Individual Silver Nanoparticles and Small Clusters*, *J. Phys. Chem. Lett.* 6, 1638–1644.

2015 Rachel Glenn, Kochise Bennett, Konstantin E. Dorfman, and Shaul Mukamel, *Photon-Exchange Induces Optical Nonlinearities in Harmonic Systems*, *J. Phys. B: At. Mol. Opt. Phys.* 48, 065401.

2014 Rachel Glenn and Shaul Mukamel, *Nonlinear Transmission Spectroscopy with Dual Frequency Combs*, *Phys. Rev. A* 90, 023804.

2014 Rachel Glenn and Shaul Mukamel, *Multidimensional Spectroscopy with a Single Broadband Phase-Shaped Laser Pulse*, The Journal of Chemical Physics 140, 144105.

2014 Rachel Glenn, Mark E. Limes, Brian Saam, Christoph Boehme, and Mikhail E. Raikh, *Analytical Study of Spin-Dependent Transition Rates within Pairs of Dipolar and Strongly Exchange Coupled Spins with* S=1/2 *During Magnetic Resonant Excitation*, *Physical Review B* 87, 165205.

2013 Rachel Glenn, Mark E. Limes, Brendan Pankovich, Brian Saam, and Mikhail E. Raikh, *Magnetic Resonance in Slowly Modulated Longitudinal Field: Modified Shape of the Rabi Oscillations*, *Physical Review B* 87, 155128.

2013 Rachel Glenn, William Jacob Baker, Christoph Boehme, and Mikhail E. Raikh, *Analytical Description of the Spin-Rabi Oscillation Controlled Electron Transition Rates Between Weakly Coupled Pairs of Paramagnetic States with* S=1/2, *Physical Review B* 87, 155208.

2012 Rachel Glenn, Oleg A. Starykh, and Mikhail E. Raikh, *Interplay of Spin-Orbit Coupling and Zeeman Splitting in the Absorption Lineshape of 2D Fermions*, *Physical Review B* 86, 024423.

2011 Rachel Glenn and Mikhail E. Raikh, *Rabi-Vibronic Resonance with Large Number of Vibrational Quanta*, *Physical Review B* 84, 195454.

PROFESSIONAL AFFILIATIONS

2015-present	American Physical Society (APS)
2021-present	American Association of Physicists in Medicine (AAPM)
2026-present	International Society for Magnetic Resonance in Medicine (ISMRM)

PROFESSIONAL PRESENTATIONS (Only presentations/posters that I had given)

Aug. 19, 2026	Quantum Computing in The Next Five Years Oral Joint NucMed Seminar at MD Anderson
Aug. 25, 2026	Mechanistic Evaluation of Diffusion MRI for Predicting Glioma Cell Density Poster CCSG Image-Guided Interventions & Insights (III) Program Retreat
Jul. 19, 2026	Mechanistic Evaluation of Diffusion MRI for Predicting Glioma Cell Density Poster American Association of Physicists in Medicine

Mar. 28, 2026	Invited Workshop Presentation and Career Panel, Reach for the stars STEM Festival Oral Rice University
Mar. 24, 2026	Quantum Algorithms for Medical Imaging: Motivation, Measurements, and Limits Oral Rice University Quantum Student Success Summit (Invited speaker)
Aug. 8, 2025	AI-Enhanced Compressed Sensing for Synthetic Aperture Radar (SAR) Imaging Oral Lockheed Martin Conference
May 15, 2024	Review of Quantum/Classical Image Processing Algorithms for Medical Images Oral Society for Thermal Medicine
Mar. 21, 2024	Development of a Segmentation Model for Liver on Magnetic Resonance Imaging Oral Tumor Measurement Initiative Symposium
Jul. 26, 2023	A Weighted Normalized Boundary Loss for Reducing the Hausdorff Distance in Medical Imaging Segmentation Poster American Association of Physicists in Medicine
Jul. 26, 2023	Quantum and Classical Image Processing Algorithms for Medical Imaging Poster American Association of Physicists in Medicine
Nov. 5, 2022	Incorporating GTV Information in a Multi-stage Process to Improve Automatically Generated Field Apertures for Rectal Cancer Radiotherapy Poster Society of Industrial and Applied Mathematics
Oct. 6, 2020	Silicon Photonics for Quantum Computing Hardware Oral Lockheed Martin Lunch and Learn
May 5, 2020	Artificial Intelligence Agent Architectures Oral

Lockheed Martin Lunch and Learn

Aug. 13, 2019 Swarming for Collaborative Agent Systems
Poster
Lockheed Martin Conference

Mar. 5, 2019 Mapping of the Excited State Potential Energy Surface during Molecular Photoisomerization to Control Chemical Reactions
Oral
American Physics Society Annual March Meeting

Nov. 5, 2018 Conceptual Insights into the Interaction of the Electric Field and Molecule in Spectroscopy to Further Our Understanding of Molecular Dynamics such as Isomerization
Oral
Lockheed Martin Lunch and Learn

Mar. 5, 2018 Conceptual Insights into the Interaction of the Electric Field and Molecule in Spectroscopy to Further Our Understanding of Molecular Dynamics Such as Isomerization
Oral
American Physics Society Annual March Meeting

Nov. 30, 2017 Introduction to Modern Quantum Chemistry and Real-Time Molecular Simulations
Oral
Lockheed Martin Lunch and Learn

Oct. 19, 2017 Review of Real-Time Molecular Simulations Results and Project Turnover
Oral
Crawford Research Group

Aug. 31, 2017 Localization Errors and Controllable Sparsity in the Coupled-Cluster Wave Function for Time- Dependent Linear Response Theory
Poster
The World Association of Theoretical and Computational Chemists 2017

Mar. 13, 2017 Time-Dependent Molecular Response with Both Electric and Magnetic Fields
Oral
American Physics Society Annual March Meeting

Aug. 19, 2016 Molecular Level-Crossing and Organic Semiconductors Microsystems
Oral
Technology Office, Booz Allen Hamilton

Aug. 3, 2016 Measuring the Real and Imaginary Parts of the Molecular Polarization using a Single-Shaped Pulse
Oral
Crawford Research Group, Virginia Polytechnic Institute and State University

Jul. 14, 2016 Molecular Level-Crossing and the Geometric Phase Effect from the Optical Hanle Perspective
Oral
Wilson Research Group, Michigan State University

Mar. 16, 2016 Molecular Level-Crossing and the Geometric Phase Effect from the Optical Hanle Perspective
Oral
American Physics Society Annual March Meeting

Jun. 30, 2015 Multidimensional Stimulated Emission Spectroscopy with a Single Phase-Shaped Pulse
Oral
The International Conference on Time-Resolved Vibrational Spectroscopy 2015

Mar. 5, 2015 Femtosecond Nanoplasmonic Dephasing of Individual Silver Nanoparticles
Oral
American Physics Society Annual March Meeting

Mar. 4, 2015 Coherent Dynamics Following Strong Field Ionization of Polyatomic Molecules
Oral
American Physics Society Annual March Meeting

Mar. 3, 2015 Solvation Dynamics with a Single Phase-Shaped Laser Pulse
Poster
American Physics Society Annual March Meeting

Apr. 4, 2014 Rabi-Vibronic Resonance and Photon-Exchange Induced Nonlinearities in Harmonic Oscillators
Oral
Physics Colloquium, University of California, Merced

Mar. 8, 2013 Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors
Oral
Naval Research Laboratory, Washington, DC

Nov. 17, 2012 Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors

Oral
Materials Research Science and Engineering Centers; Engineering/Physics Department, University of Utah

Oct. 4, 2012 Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors
Oral
Quantum Information Processing; Physics Department, University of Wisconsin-Madison

Mar. 20, 2012 Rabi-Vibronic Resonance with a Large Number of Vibrational Quanta
Oral
Physics Department, University of Utah

Jan. 24, 2012 Peculiarities of the Rabi Oscillations in a Quantum Dot Molecule
Oral
Physics Department, University of Utah